

NAIVE Bayes classifier

2-class, C_0 and C_1

Full PDFs or PMFs

$$\left\{ \begin{array}{l} P(x|C_0) \equiv \text{data likelihood for } C_0 \\ P(x|C_1) \equiv \text{" " " } C_1 \end{array} \right.$$

numeric values

$$\left\{ \begin{array}{l} P(C_0) \equiv \text{prior prob. for } C_0 \\ P(C_1) \equiv \text{" " " } C_1 \end{array} \right.$$

For each $P(x | C_i)$, $i = 0, 1$,
we must solve for its
parameters. Using MLE
or MAP.

e.g. if $P(x | C_0) \sim G(x | \mu_0, \sigma_0^2)$

using MLE: $\mathcal{L}^0 = \left(\prod_{i=1}^N G(x_i | \mu_0, \sigma_0^2) \right)$

and then take $\mathcal{L} = \ln \mathcal{L}^0$

and derive for μ_0, σ_0^2

$$\text{i.e. } \frac{\partial \mathcal{L}}{\partial \mu_0} = 0 \text{ and } \frac{\partial \mathcal{L}}{\partial \sigma_0^2} =$$

Using MAB:

$$\textcircled{1} \mathcal{L}^0 = \left(\prod_{i=1}^N G(x_i | \mu_0, \sigma_0^2) \right) \cdot P(\mu_0) \cdot P(\sigma_0^2)$$

Gaussian

Laplacian

$$\textcircled{2} \mathcal{L} = \ln(\mathcal{L}^0)$$

$$\textcircled{3} \frac{\partial \mathcal{L}}{\partial \mu_0} = 0 \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \sigma_0^2} = 0.$$

Gaussian Mixture Models

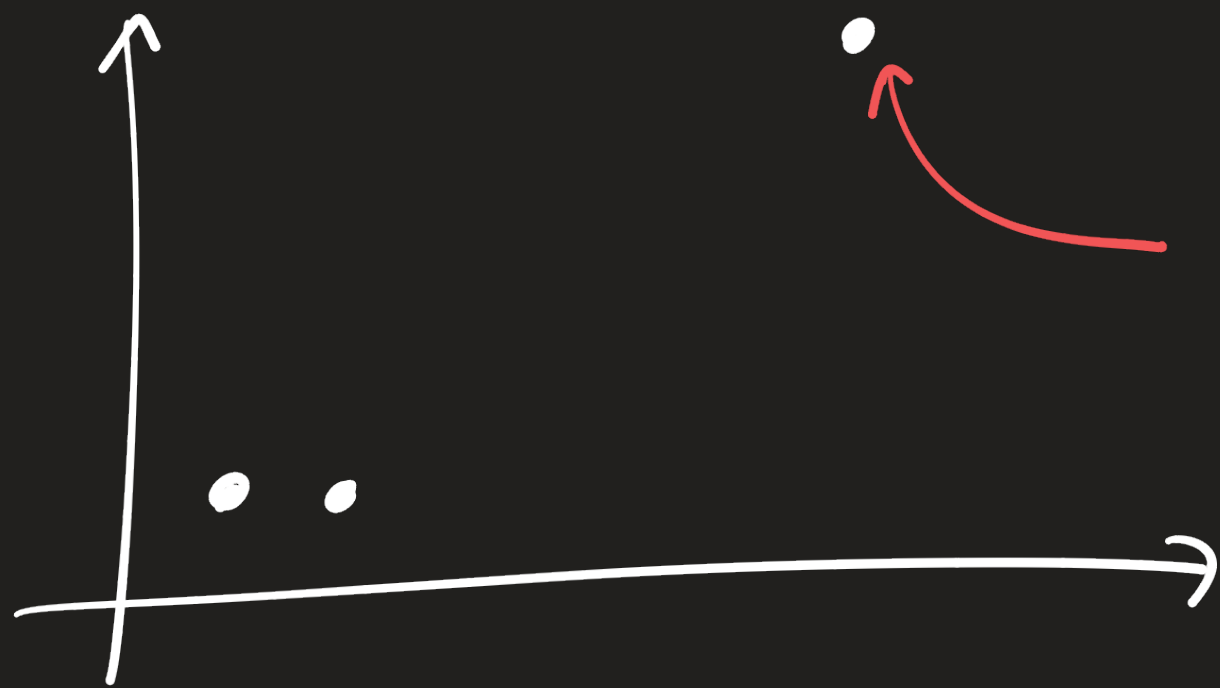
$$P(x|\theta) = \sum_{k=1}^K \pi_k \cdot G(x|\mu_k, \Sigma_k)$$

and $0 \leq \pi_k \leq 1$

and $\sum_{k=1}^K \pi_k = 1$

$$\mathcal{L}^o = \prod_{i=1}^N \sum_{k=1}^K \pi_k \cdot G(x_i|\mu_k, \Sigma_k)$$

intractable data
likelihood.



each point is
sampled by
only one component.

EM introduces this hidden non-observable
variable (AKA hidden latent variables)
and estimates it (E-step).

Latent Variables

$z_i \equiv$ component label
from which sample
 x_i was drawn
from.

$$\underline{z_i \in \{1, 2, \dots, K\}}$$

MEMBERSHIP
MATRIX
All z_i 's $[z_i]$

$$\begin{matrix} C \\ \uparrow \\ N \end{matrix} = \begin{bmatrix} \\ \\ \\ \end{bmatrix} \rightarrow Z = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

$$\mathcal{L}^c = \prod_{i=1}^N \pi_{z_i} \cdot G(x_i | \mu_{z_i}, \Sigma_{z_i})$$

MAP for solving for $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$

$$\textcircled{1} \mathcal{L}^q = \left(\prod_{i=1}^N \pi_{z_i} G(x | \mu_{z_i}, \Sigma_{z_i}) \right) \cdot \left(\prod_{z_i=1}^K G(\mu_{z_i} | \mu_0, \Sigma_0) \right)$$

$$\textcircled{2} \mathcal{L} = \ln(\mathcal{L}^q)$$

$$\textcircled{3} \frac{\partial \mathcal{L}}{\partial \mu_{z_i}} = 0, \quad \frac{\partial \mathcal{L}}{\partial \pi_{z_i}} = 0, \quad \frac{\partial \mathcal{L}}{\partial \Sigma_{z_i}} = 0$$